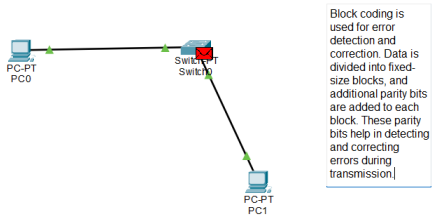
Experiment 4: Error Detection using CRC and Block Coding

Aim

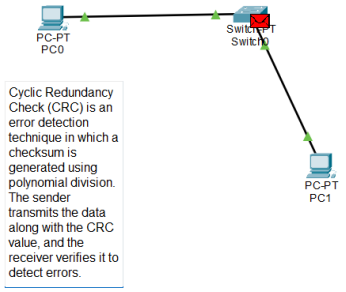
To understand and analyze error detection techniques such as **Cyclic Redundancy Check (CRC)** and **Block Coding** in data communication.

Theory

Block Coding:

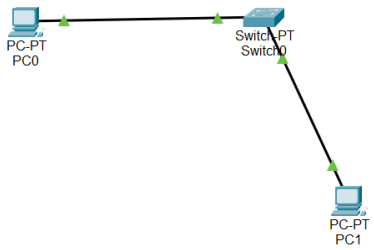


Cyclic Redundancy Check (CRC):



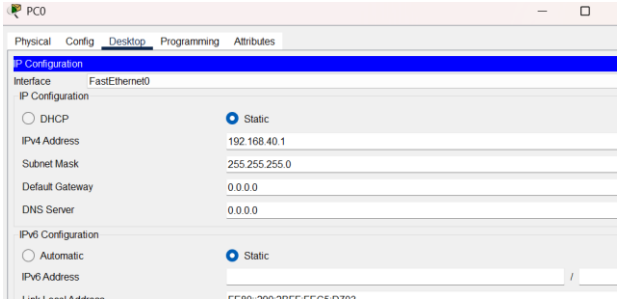
Network Topology

- Connect both PCs to a switch



Step 1: IP Configuration

- PC0 → 192.168.40.1
- PC1 → 192.168.40.2
- Subnet Mask → 255.255.255.0

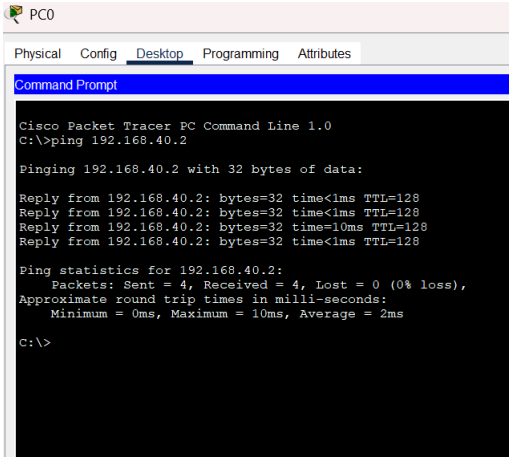


Step 2: Connectivity Test

On PC0:

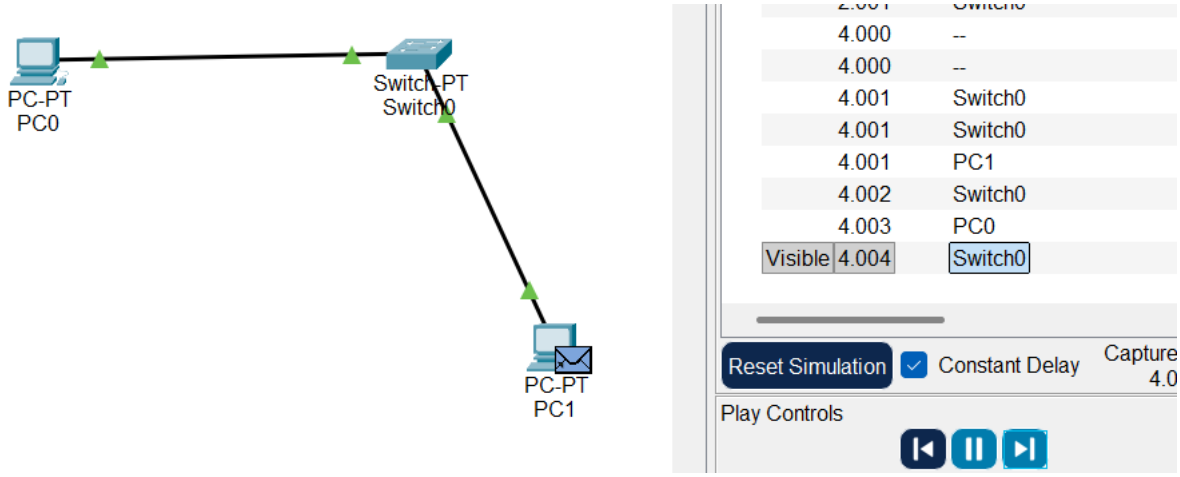
ping 192.168.40.2

- Successful replies are observed



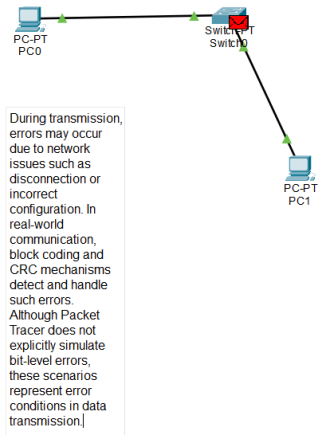
Step 3: Simulation Mode

- Switch to Simulation Mode
- Use “Add Simple PDU”
- Send packet from PC1 to PC0
- Observe packet flow through switch



Observations

- Data is successfully transmitted between PCs
- Packet travels through switch without error
- No visible corruption in data
- Error detection occurs internally during transmission



Result

CRC and block coding are important error detection techniques used in data communication. Although Cisco Packet Tracer does not explicitly display these processes, they ensure reliable data transmission by detecting errors in transmitted data.